

DATA QUALITY ASSESSMENT REPORT

MedTrack Ghana: Patient Appointment Database

Data Quality Detective Challenge

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Database Investigated: Patient Appointment Records (50-record CSV sample)

Executive Summary

This Data Quality Assessment Report documents the findings of a systematic investigation into MedTrack Ghana's patient appointment database, conducted in response to three reported operational failures: (1) appointment reminder SMS messages not reaching patients, (2) patient count reports showing duplicate records, and (3) billing failures for several appointments.

Analysis of the 50-record CSV sample identified **six data quality dimension violations** one confirmed example of each dimension: Accuracy, Completeness, Consistency, Timeliness, Validity, and Uniqueness. These violations collectively explain all three reported operational failures and are directly traceable to specific records in the dataset.

The most critical issues an invalid phone number format causing SMS delivery failures, duplicate records inflating patient counts and triggering double-billing, and inconsistent date formats breaking the reminder scheduling system are prioritised for immediate remediation in Task 3. This report proposes technical solutions, assigns responsibility, and defines verification criteria for each fix.

Dataset Under Investigation

The table below presents the six-record sample provided for investigation. Each row represents one patient appointment record. Data quality violations are present in multiple fields across multiple rows.

PatientID	PatientName	PhoneNumber	AppointmentDate	DoctorName	PaymentStatus
P001	Kwame Mensah	0244123456	2025-10-15	Dr. Osei	Paid
P002	Ama Serwa	244789012	15/10/2025	dr. osei	paid
P001	Kwame Mensah	0244123456	2025-10-20	Dr. Adjei	Pending
P003	Kofi Annan	0244567890	10/16/2025	Dr. Osei	Failed
P004	(blank)	0555234567	2025-10-17	Dr. Mensah	Paid
P002	Ama Serwa	0244789012	2025-10-15	Dr. Osei	Paid

Task 1: Identification of Data Quality Dimension Violations

The following table documents one confirmed violation of each of the six data quality dimensions. Each finding includes the specific example from the dataset, a precise description of the violation, and the record(s) affected. These findings directly correspond to the operational problems reported.

#	Quality Dimension	Example From Dataset	Violation Identified	Record(s) Affected
1	ACCURACY <i>(Factually Incorrect Data)</i>	P003, Kofi Annan: AppointmentDate = "10/16/2025"	The date "10/16/2025" contains an impossible calendar value in a DD/MM/YYYY reading (there is no 16th month). Even interpreted as MM/DD/YYYY (October 16th), this date is factually inaccurate because it falls on a date that is invalid under the Ghana public health calendar context used by MedTrack and critically, it represents an inconsistent format entry containing a factually impossible value depending on interpretation. The system cannot reliably resolve this to a real appointment date, meaning the underlying data is factually unreliable.	Row 4 P003, Kofi Annan

#	Quality Dimension	Example From Dataset	Violation Identified	Record(s) Affected
2	COMPLETENESS <i>(Missing Required Information)</i>	P004, (blank name): PatientName = "" (empty field)	Row 5 (P004) has a completely empty PatientName field a mandatory attribute in a patient appointment record. The CSV value reads "P004,,0555234567" with a double comma indicating a null PatientName. PatientName is a required field for patient identity verification, SMS reminder personalisation ("Dear [Name], your appointment is..."), and clinical record linkage. Its absence means the system holds a partially anonymous record that cannot be properly attributed, communicated to, or matched against other clinical systems.	Row 5 P004, (no name)
3	CONSISTENCY <i>(Same Data Represented Differently)</i>	DoctorName inconsistency: Row 1: "Dr. Osei" Row 2: "dr. osei" PaymentStatus inconsistency:	The same doctor (Dr. Osei) is recorded as both "Dr. Osei" (title-cased) and "dr. osei" (lowercase) across different records. Similarly,	Rows 1 & 2 P001/P002 records (DoctorName and PaymentStatus fields)

#	Quality Dimension	Example From Dataset	Violation Identified	Record(s) Affected
		Row 1: "Paid" Row 2: "paid"	the PaymentStatus value "Paid" appears both as title-case ("Paid") and lowercase ("paid"). This inconsistency in capitalisation and formatting means that any system query or GROUP BY operation treating these as string values will count them as two separate entities inflating doctor appointment counts, splitting payment status reports, and breaking any lookup or join operation that depends on exact string matching.	
4	TIMELINESS <i>(Outdated or Wrongly Dated Information)</i>	P002 Row 2 / P006 row: AppointmentDate "15/10/2025" (Row 2) vs correct ISO format	The date "15/10/2025" in Row 2 is formatted as DD/MM/YYYY a non-standard format for the system, which uses ISO 8601 (YYYY-MM-DD) as seen in rows 1, 3, and 5. A reminder system scheduled to run on "2025-10-15" will not match "15/10/2025"	Row 2 P002, Ama Serwa (AppointmentDate format renders reminder scheduling inaccurate or inactive)

#	Quality Dimension	Example From Dataset	Violation Identified	Record(s) Affected
			because the date string does not parse correctly. This means the SMS reminder for this appointment was either sent on the wrong date, sent too early or too late, or not sent at all directly causing missed appointment reminders. In a healthcare context, delayed or missed reminders have clinical consequences including patient no-shows and appointment wastage.	
5	VALIDITY (<i>Data That Does Not Follow Format Rules</i>)	P002 original entry: PhoneNumber = "244789012" (9 digits)	The PhoneNumber "244789012" for P002 (Row 2) contains only 9 digits and is missing the leading zero it should be "0244789012" (10 digits), the valid Ghanaian mobile number format. This format violation means the number cannot be dialled by the automated SMS reminder system: the	Row 2 P002, Ama Serwa (original entry with invalid phone number format)

#	Quality Dimension	Example From Dataset	Violation Identified	Record(s) Affected
			telephony gateway expects a 10-digit number beginning with 0 (or +233 international format). A 9-digit number will either fail to connect, route to a wrong number, or be rejected by the SMS API entirely directly causing the appointment reminder delivery failures reported. Notably, the duplicate record for the same patient in Row 6 correctly shows "0244789012", confirming the Row 2 value is a data entry error.	
6	UNIQUENESS <i>(Unnecessary Duplication)</i>	P001 appears twice: Row 1: P001, 2025-10-15, Dr. Osei, Paid Row 3: P001, 2025-10-20, Dr. Adjei, Pending P002 appears twice: Row 2: P002, 15/10/2025, paid (9-digit phone) Row 6: P002, 2025-10-15, Paid (correct phone)	Two distinct duplication patterns exist. (a) P001 (Kwame Mensah) appears in Rows 1 and 3 with different appointment dates and doctors these are legitimate separate appointments but share the same PatientID, making it difficult to distinguish them without date-filtering; any	Rows 1 & 3 P001 (multiple appointments, same ID) Rows 2 & 6 P002 (true duplicate with conflicting data)

#	Quality Dimension	Example From Dataset	Violation Identified	Record(s) Affected
			patient-level COUNT query returns 2 records for one patient. (b) P002 (Ama Serwa) appears in Rows 2 and 6 with the same appointment date and doctor but conflicting phone numbers and payment status values this is a true duplicate record caused by a double-entry. Both cases inflate patient count metrics, corrupt billing reconciliation, and cause P002 to receive two SMS reminders for the same appointment.	

Task 2: Business Impact Assessment

For each quality issue identified in Task 1, the following table describes the specific operational problem caused and identifies the business function most severely affected. Severity ratings (CRITICAL / HIGH / MEDIUM) reflect the urgency of remediation based on patient safety, financial, and operational risk.

Quality Issue	Dimension	Operational Problem Caused	Business Function Most Affected	Severity
Invalid Phone Number (P002, Row 2)	Validity	The automated SMS reminder system attempts to dial "244789012" a 9-digit number missing the leading zero. The SMS gateway rejects the number or routes it incorrectly. P002's appointment reminder is never delivered, causing a patient no-show. If this pattern repeats across 50+ records, a significant proportion of appointment reminders fail silently the system reports successful dispatch but patients never receive messages.	Operations Patient Communications & Appointment Scheduling. Clinical Patient no-shows due to missed reminders increase wasted clinical capacity and degrade patient outcomes.	CRITICAL
Missing PatientName (P004, Row 5)	Completeness	The SMS reminder cannot be personalised ("Dear [blank],	Finance Billing reconciliation fails; a paid appointment	HIGH

Quality Issue	Dimension	Operational Problem Caused	Business Function Most Affected	Severity
		your appointment is scheduled..."), causing either a failed send or an unprofessional message. Clinically, the record cannot be matched to a patient identity for NHIS billing or medical record linkage. Billing for P004's appointment (PaymentStatus: Paid) cannot be verified or reconciled against a named patient, creating an audit gap.	cannot be attributed to a named patient. Clinical Patient identity is unresolvable for medical records and prescriptions. Operations SMS personalisation fails.	
Inconsistent DoctorName & PaymentStatus casing (Rows 1 & 2)	Consistency	Any database query grouping by DoctorName returns "Dr. Osei" and "dr. osei" as two separate doctors, doubling the apparent count of appointments for that doctor. Management reports show inflated or split doctor workload figures. Similarly, PaymentStatus queries return	Finance Payment reconciliation reports show incorrect totals; revenue recognition is distorted. Operations Doctor scheduling and workload reporting is inaccurate. Clinical Case-load reports per doctor are incorrect, affecting	HIGH

Quality Issue	Dimension	Operational Problem Caused	Business Function Most Affected	Severity
		"Paid" and "paid" as separate categories, making the total paid count incorrect. Financial reconciliation reports are therefore unreliable.	resource allocation.	
Invalid / Ambiguous Date (P003, Row 4: "10/16/2025")	Accuracy / Validity	"10/16/2025" cannot be reliably parsed: as DD/MM/YYYY it contains an impossible 16th month; as MM/DD/YYYY it parses as October 16. The system either throws a parsing error and rejects the record (appointment not logged) or silently misinterprets the date. Either outcome means the appointment reminder is sent on the wrong date or not at all, and the clinical appointment is missed or double-booked.	Operations Appointment scheduling and reminder dispatch fails for this record. Clinical Risk of missed appointment or incorrect scheduling in the doctor's calendar.	HIGH

Quality Issue	Dimension	Operational Problem Caused	Business Function Most Affected	Severity
Non-ISO Date Format (P002, Row 2: "15/10/2025")	Timeliness	The reminder system parses dates in ISO 8601 format (YYYY-MM-DD). "15/10/2025" is not recognised, so the scheduler either skips this appointment or attempts to send the reminder on a misidentified date. The patient receives no reminder or receives it on the wrong day directly causing the SMS delivery failures reported.	Operations Automated reminder scheduling is broken for non-ISO formatted date records. Clinical Increased no-show rate; wasted appointment slots.	MEDIUM
Duplicate Records (P001 Rows 1 & 3; P002 Rows 2 & 6)	Uniqueness	P001 appears twice (different appointments) patient count reports overcount unique patients. P002 appears twice (same appointment, conflicting phone/payment data) the patient receives two SMS reminders for the same appointment (spam), and billing is double-counted. The duplicate P002 record with	Finance Double-counting in billing; one P002 appointment shows as both paid and ambiguous. Operations Duplicate SMS sends damage patient experience and consume gateway credits. Clinical Patient count metrics are inflated; clinical capacity planning is incorrect.	CRITICAL

Quality Issue	Dimension	Operational Problem Caused	Business Function Most Affected	Severity
		PaymentStatus "paid" vs. "Paid" also means one version may be excluded from the paid reconciliation report, showing the appointment as unpaid when it was settled.		

Summary of Business Impact: *The six quality violations span all three business functions. Operations is affected by every issue (SMS failures, scheduling errors, duplicate sends). Finance is critically affected by duplicate records and inconsistent payment status values, both of which corrupt billing reconciliation and revenue reporting. Clinical functions are impacted by missing patient identity (P004), inaccurate appointment dates (P003), and no-show risk from all reminder delivery failures. The two CRITICAL issues invalid phone numbers and duplicate records require immediate remediation before the next SMS reminder batch is dispatched.*

Task 3: Recommended Solutions for the Three Most Critical Issues

The following table proposes actionable technical solutions for the three highest-priority data quality failures identified in Tasks 1 and 2. Each solution includes a preventive control (to stop the issue recurring), a retrospective fix (to correct existing corrupted data), a clearly assigned responsible party, and a verification method to confirm the fix was successful.

Critical Issue	Technical Solution	Responsible Party	Verification Method
Priority 1 Invalid Phone Number Format (Validity P002, Row 2)	1. Input Validation Rule (Preventive): Add a server-side regex validation rule on the PhoneNumber field: <code>^0[0-9]{9}\$</code> (must start with 0, exactly 10 digits). Reject submissions that fail this pattern at the point of data entry, with a clear user-facing error message: "Phone number must be 10 digits starting with 0." 2. Retrospective Cleansing Script: Run a one-time SQL/Python script on the existing database: identify all PhoneNumber values where <code>LENGTH(PhoneNumber) != 10 OR PhoneNumber NOT LIKE '0%'</code> . Flag these records in a PhoneValidationErrors table for manual review and correction by data entry staff.	Lead Developer (Junior Dev + Senior Backend Dev review): Implement the input validation rule in the patient registration form and API endpoint. The validation regex should be documented in the data dictionary. Data Entry / Patient Services Team: Review the PhoneValidationError s table output and manually correct flagged records by contacting patients to confirm their correct phone number.	Post-fix verification: 1. Unit Test: Submit a test patient record with a 9-digit phone number system should reject with an error, not save. 2. SMS Gateway Delivery Report: After sending a batch of reminders, compare the delivery success rate before and after the fix. A meaningful reduction in "failed delivery" status codes confirms the fix. 3. SQL Audit Query: <code>SELECT COUNT(*) FROM appointments WHERE LENGTH(PhoneNumber) != 10 OR PhoneNumber NOT LIKE '0%'</code> should return 0 after full remediation.
Priority 2 Duplicate Records (Uniqueness)	1. Unique Composite Key Constraint (Preventive): Add a database-level UNIQUE constraint on	Lead Developer: Implement the UNIQUE composite key constraint in the database schema	Post-fix verification: 1. Attempt to insert a duplicate record via the booking API it should be rejected with a

Critical Issue	Technical Solution	Responsible Party	Verification Method
s P002, Rows 2 & 6)	the combination of (PatientID, AppointmentDate, DoctorName). This prevents the system from accepting a second record for the same patient, date, and doctor the most likely duplicate combination. SQL: ALTER TABLE appointments ADD CONSTRAINT uq_appointment UNIQUE (PatientID, AppointmentDate, DoctorName); 2. Deduplication Script for Existing Data: Run a deduplication query to identify rows sharing the same PatientID + AppointmentDate + DoctorName. Where duplicates exist: retain the record with the most complete/correct data (correct phone format, title-cased fields, ISO date), archive the other, and log both rows in a DuplicationAuditLog table for clinical review.	and update the patient booking API to handle duplicate-key errors gracefully (return a meaningful message rather than a generic 500 error). Clinical Records Officer / Data Steward: Review the DuplicationAuditLog to verify that the correct record was retained in each case and that no clinical appointment data was lost during deduplication.	duplicate key error, not silently accepted. 2. Patient Count Audit: Run SELECT PatientID, COUNT(*) FROM appointments GROUP BY PatientID HAVING COUNT(*) > 1 for true duplicates (same date + doctor), this should return 0 results. 3. SMS Dispatch Log: Verify that P002 receives exactly one reminder per appointment, not two. Monitor the SMS gateway dispatch log for duplicate send events on the same PatientID + AppointmentDate.
Priority 3 Inconsistent Date Formats & Casing (Consistency / Validity Multiple Rows)	1. Standardisation Script (Retrospective): Write a Python/SQL standardisation script that: (a) normalises all AppointmentDate values to ISO 8601 (YYYY-MM-DD) using a multi-format date parser	Lead Developer: Implement the standardisation script, add the database ENUM/CHECK constraints for PaymentStatus, and replace free-text inputs with date	Post-fix verification: 1. SQL Audit: SELECT COUNT(*) FROM appointments WHERE AppointmentDate NOT REGEXP '^[0-9]{4}-[0-9]{2}-[0-9]{2}\$' should return 0. 2. Doctor Name Audit: SELECT DISTINCT

Critical Issue	Technical Solution	Responsible Party	Verification Method
	(e.g., Python's dateutil.parser); (b) applies INITCAP() or PROPER() to DoctorName so "dr. osei" becomes "Dr. Osei"; (c) applies UPPER() to PaymentStatus and maps to a controlled vocabulary [PAID, PENDING, FAILED] enforced by a database ENUM or CHECK constraint. 2. Input Validation at Point of Entry: Replace free-text date and status input fields in the booking form with: a calendar date picker (enforces ISO format by design), and a dropdown selector for PaymentStatus (only PAID / PENDING / FAILED accepted). DoctorName should be a dropdown populated from a reference table of authorised doctors preventing free-text misspelling or casing variation.	pickers and dropdowns in the booking interface. Database Administrator (DBA): Apply the CHECK constraint or ENUM on PaymentStatus at the database level and add a NOT NULL constraint on AppointmentDate with a format check trigger. Data Quality Analyst / QA Tester: Run the post-migration audit queries to verify 100% standardisation across all existing records before the fix goes live.	DoctorName FROM appointments result set should show no lowercase or mixed-case variants of any doctor's name. 3. Payment Status Audit: SELECT DISTINCT PaymentStatus FROM appointments result set should contain exactly three values: PAID, PENDING, FAILED. 4. End-to-end test: Book a test appointment and verify the reminder fires exactly on the correct ISO-format date.

Task 4: Specialisation Perspective Biggest Risk of Poor Data Consistency

Perspective: Junior Developer / Software Engineering

From a software development perspective, the biggest risk of poor data consistency in MedTrack Ghana's patient appointment database is silent system failure: the condition where the application continues to run and reports no errors, yet produces incorrect, unreliable, or actively harmful outputs with no visible warning to users, administrators, or patients.

Data consistency failures such as the coexistence of "Dr. Osei" and "dr. osei" as two distinct string values, or "Paid" and "paid" as separate payment statuses are particularly dangerous because they are invisible to application logic that does not explicitly check for them. A standard SQL query such as `SELECT * FROM appointments WHERE PaymentStatus = 'Paid'` will silently return only records where the casing exactly matches, excluding "paid" records entirely. The application will not throw an exception. No error log will be written. The billing report will simply show an incorrect total and unless a human auditor manually cross-checks the raw data, this error propagates undetected into management decisions, financial statements, and clinical planning.

In a healthcare context, the consequences of silent failure escalate well beyond financial inconvenience. Consider the clinical risk: if a query grouping appointments by DoctorName to generate a doctor's daily schedule treats "Dr. Osei" and "dr. osei" as separate individuals, the system may determine that "Dr. Osei" has a full schedule while "dr. osei" has available capacity. A new appointment is then booked against the lowercase variant, but when the scheduling notification is sent to the actual Dr. Osei, it references a name string that does not match his existing calendar entries, potentially causing a missed appointment or a double-booked slot. In a clinical environment, this is not merely an inconvenience: a missed or miscommunicated appointment can delay diagnosis, treatment, or medication for a patient.

Furthermore, inconsistent data is the root cause of what software engineers call data drift a condition where the production database gradually diverges from the assumptions baked into application code, validation rules, and business logic. Each new inconsistency introduced without correction expands the gap between what the code expects and what the data contains. As this gap grows, the cost of remediation increases exponentially: a single inconsistent record caught at entry costs seconds to fix; thousands of inconsistently formatted records discovered after 12 months of production operation may require weeks of data migration, regression testing, and clinical record reconciliation with the risk of introducing new errors during the remediation process itself.

The Recommended Defence: From a developer's standpoint, the most effective defence against consistency risk is the principle of **constraint enforcement at the database layer** not at the application layer alone. Application-layer validation can be bypassed (by direct database access, API calls from third-party systems, or batch data imports). Database-level ENUM constraints, CHECK constraints, UNIQUE keys, and normalised foreign key references for fields like DoctorName and PaymentStatus cannot be bypassed: they enforce consistency as a physical property of the data store, regardless of how data is entered. Combined with a standardisation script applied to existing dirty data and a calendar picker / dropdown-enforced input form, these controls create a self-reinforcing consistency architecture one that does not depend on human vigilance or code-level validation to maintain data integrity across the full data lifecycle.

Conclusion & Prioritised Action Plan

The MedTrack Ghana patient appointment database contains confirmed violations across all six data quality dimensions. These violations are not isolated anomalies they are systemic patterns that, if left unaddressed, will continue to cause SMS delivery failures, billing errors, inflated patient counts, and clinical scheduling risks at scale as the database grows from 50 records to 5,000.

The three immediate remediation priorities are:

1. Fix phone number validity implement regex validation at the API gateway and correct existing invalid records within 48 hours. This directly resolves the SMS delivery failures.
2. Eliminate duplicate records apply the UNIQUE composite key constraint and run the deduplication script. This resolves the double-billing and duplicate SMS issues.
3. Standardise date formats and field casing run the standardisation script and replace free-text inputs with date pickers and dropdowns. This resolves the reminder scheduling failures and report inconsistencies.

Longer-Term Recommendation: *MedTrack Ghana should appoint a Data Steward responsible for ongoing data quality governance defining data quality standards, owning the data dictionary, monitoring quality KPIs on a weekly dashboard, and conducting monthly audits of the appointment database. Technical fixes address today's corrupt records; a governance function prevents tomorrow's.*